Chapter 5
Goods and financial markets: the IS-LM model

IS relation from chapter 3

Remember from chapter 3:

- Equilibrium in the goods market: production is equal to demand

$$Y = Z$$

- This condition is called the ***IS* relation**

- In the simple model of Chapter 3, the **interest rate did not affect the demand for goods**

- In fact, the equilibrium condition was given by

$$Y = C(Y - T) + \bar{I} + G$$

…no role for the interest rate!

Investment function

Investment was **constant!**

- Does this make sense?
- What does investment depend on?

Investment depends primarily on two factors:

- **The level of sales or production (+)**
  The higher are sales or production Y, the more firms will buy investment goods

- **The interest rate (-)**
  To buy investment goods firms typically have to borrow. The lower is *i*, the lower is the cost of borrowing, and the higher is investment, for given *Y*

- Using math: $I = I \underset{(+,-)}{(Y, i)}$

IS relation

- The condition for equilibrium in the goods markers becomes

$$Y = C(Y - T) + I(Y, i) + G$$

- This is the (expanded) IS relation: **a negative relation between output *Y* and the interest rate *i***, as we will see

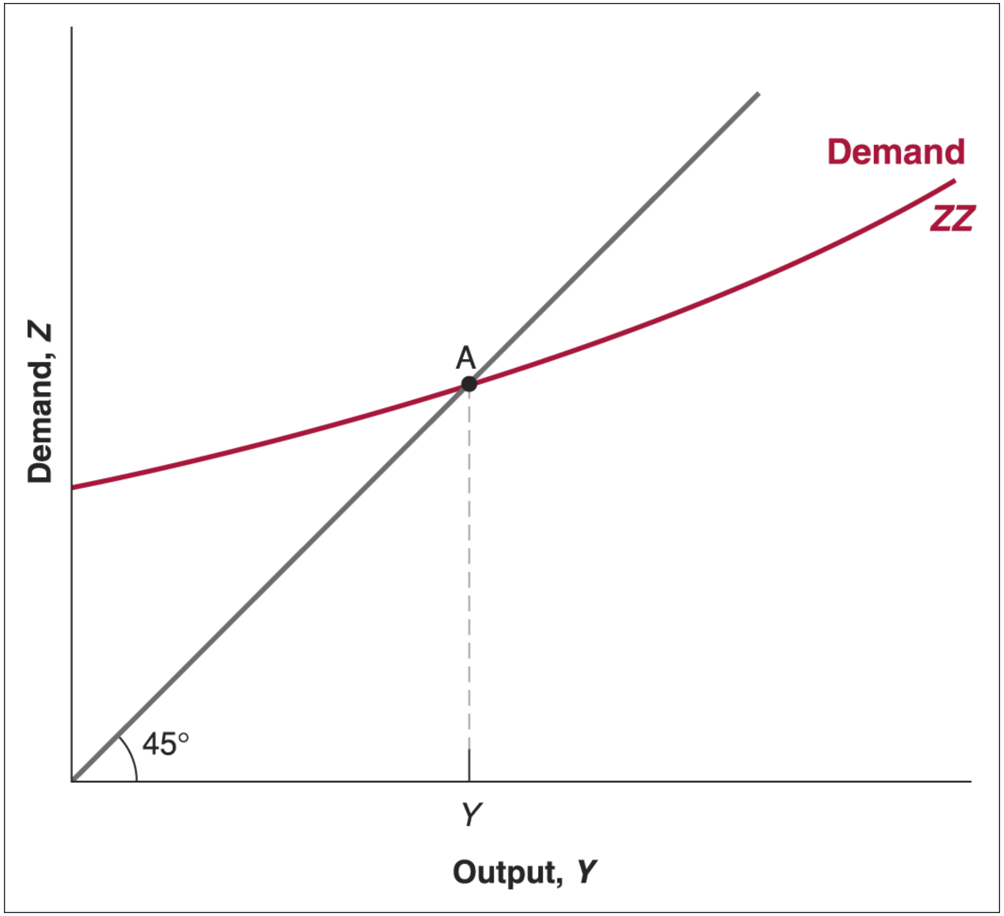
$$Y = C(\underset{+}{Y - T}) + I(\underset{+}{Y}, \underset{-}{i}) + G$$

First, note that now **demand** is an **increasing** function of **Y**, for **two reasons**:

- An increase in output leads to an increase in income and, thus, to an increase in **consumption**

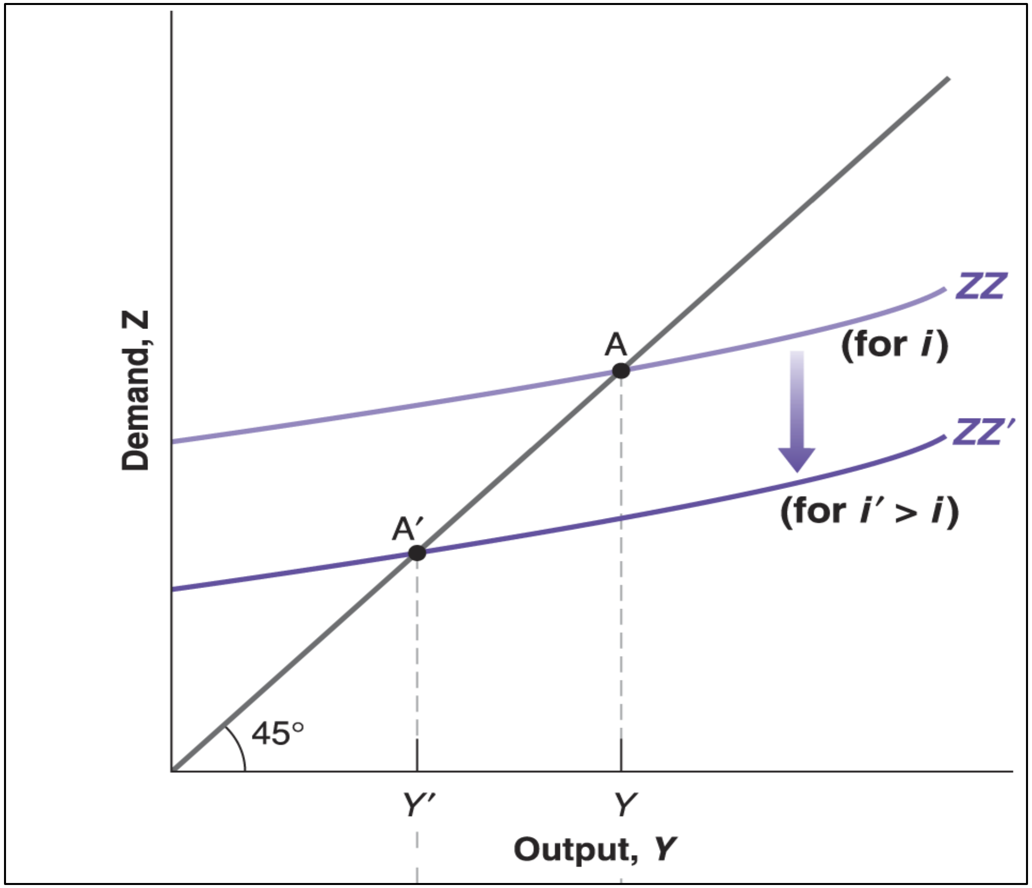
- An increase in output also leads to an increase in **investment**

How does the new graph look like?



- The demand curve *ZZ* continues to be upward sloping, with slope less than 1
- Since we have not specified the investment function, we draw *ZZ* as a curve as opposed to a line

What happens if the interest rate increases?



- An increase in the interest rate *i*, decreases investment *I* for given output *Y*, thus decreases the demand for goods for given output *Y*, therefore the demand curve *ZZ* shifts down
- In the new goods markets equilibrium output decreases from *Y* to *Y'*

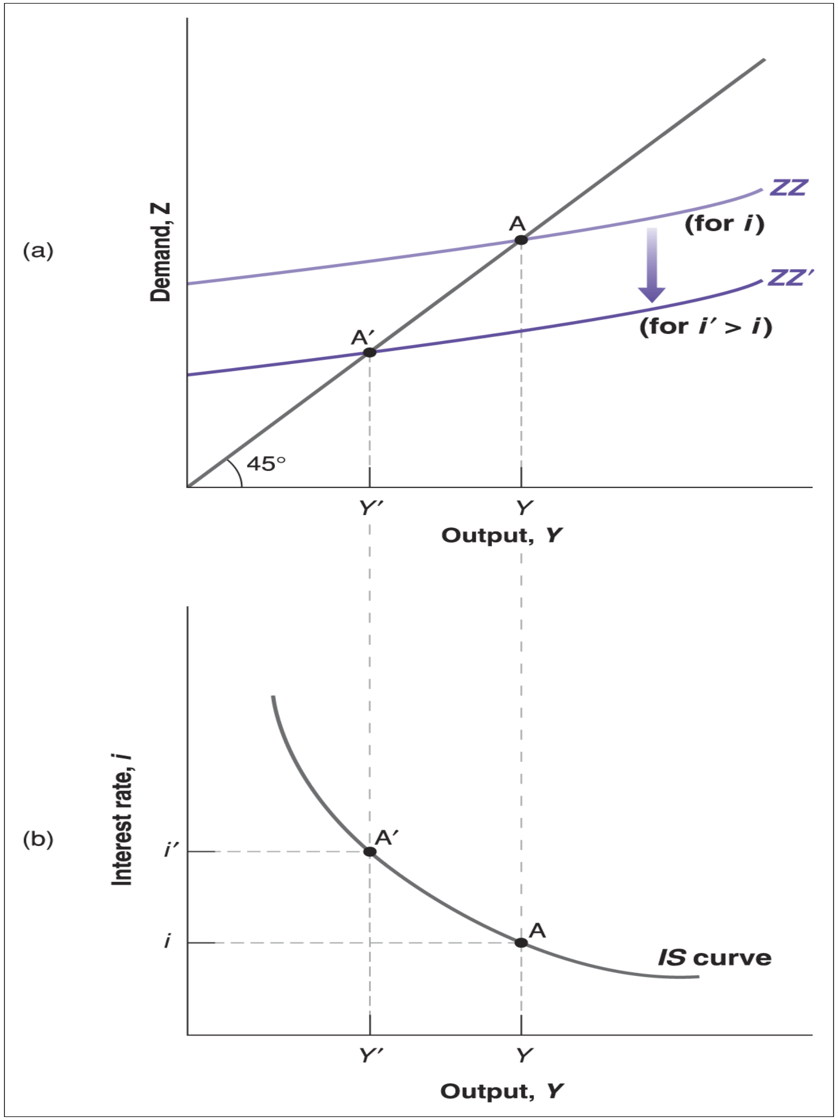
Deriving the IS curve

a) The IS curve represents the goods market equilibrium in the plan (i, Y)

b) We saw that an increase in the interest rate, decreases the demand for goods, therefore equilibrium Y decreases

c) An increase in the interest rate thus leads to a decrease in output: the **IS curve** is therefore **downward sloping**

d) The IS curve represents all the **combinations of output *Y* and interest rate *i*** for which the **goods market** is in **equilibrium**



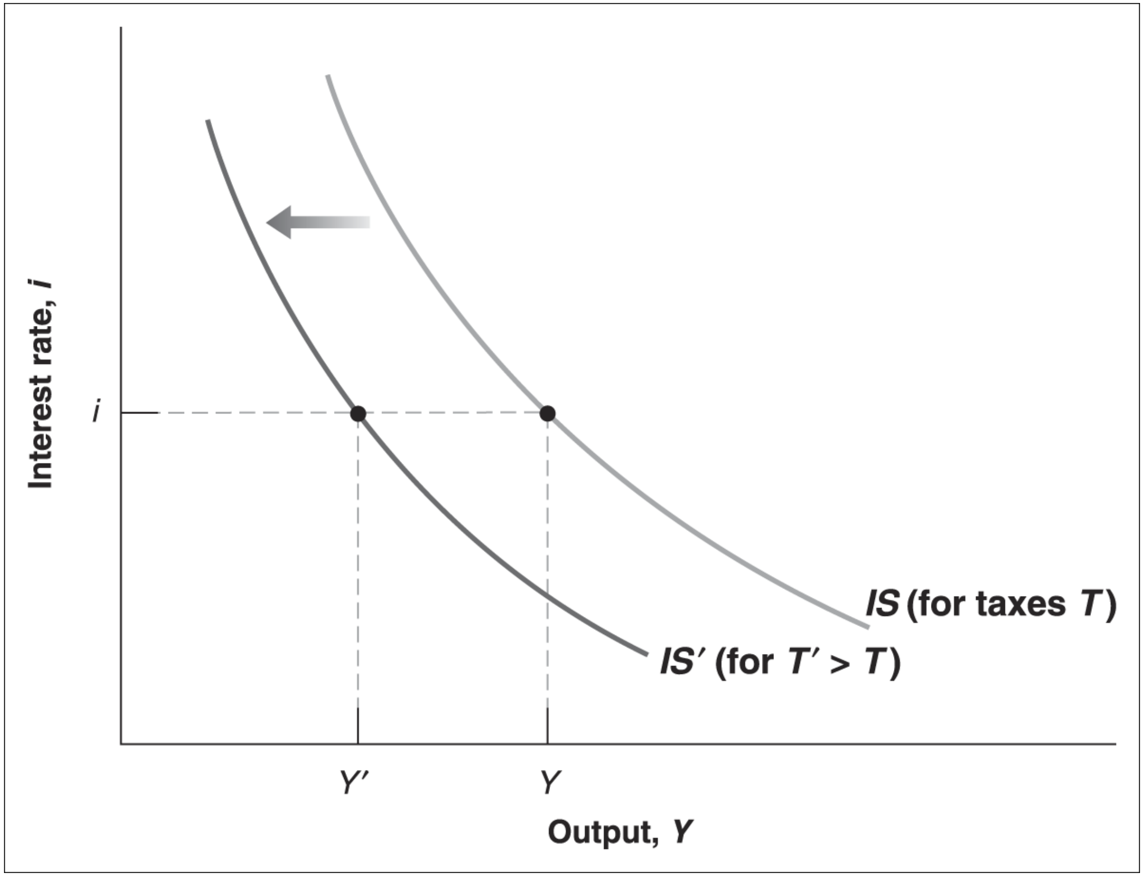
Shifts in the IS curve

- Changes in T and G will shift the IS curve!

- For example, an **increase in** taxes **T** for given interest rate decreases disposable income, decreases consumption, decreases demand and output and **shifts the IS curve to the left**

- Similarly, a **decrease in** government spending **G** or a **decrease in** consumer confidence **c_0 shift the IS curve to the left**

Example: higher T or lower G or lower c_0



To summarize:

- An **increase in the interest rate** leads to a **decrease in output** and correspond to **movements along the downward sloping IS curve**

- Changes in factors that **decrease the demand for goods**, for given interest rate, **shift the *IS* curve to the left** (for example increase in T, decrease in G, decrease in c_0)

- Changes in factors that **increase the demand for goods**, for given interest rate, **shift the *IS* curve to the right** (for example decrease in T, increase in G, increase in c_0)

LM relation

Remember from chapter 4:

- Equilibrium in financial markets: money supply is equal to money demand

$$M = €YL(i)$$

- Recall that

$$\text{Nominal GDP} = \text{Real GDP} \times P$$

$$€Y = Y \times P$$

- We can rewrite the equilibrium condition in financial markets as

$$M = Y\ P\ L(i)$$

$$M/P = YL(i)$$

Two ways of writing the LM relation

- In terms of nominal demand of money and nominal income

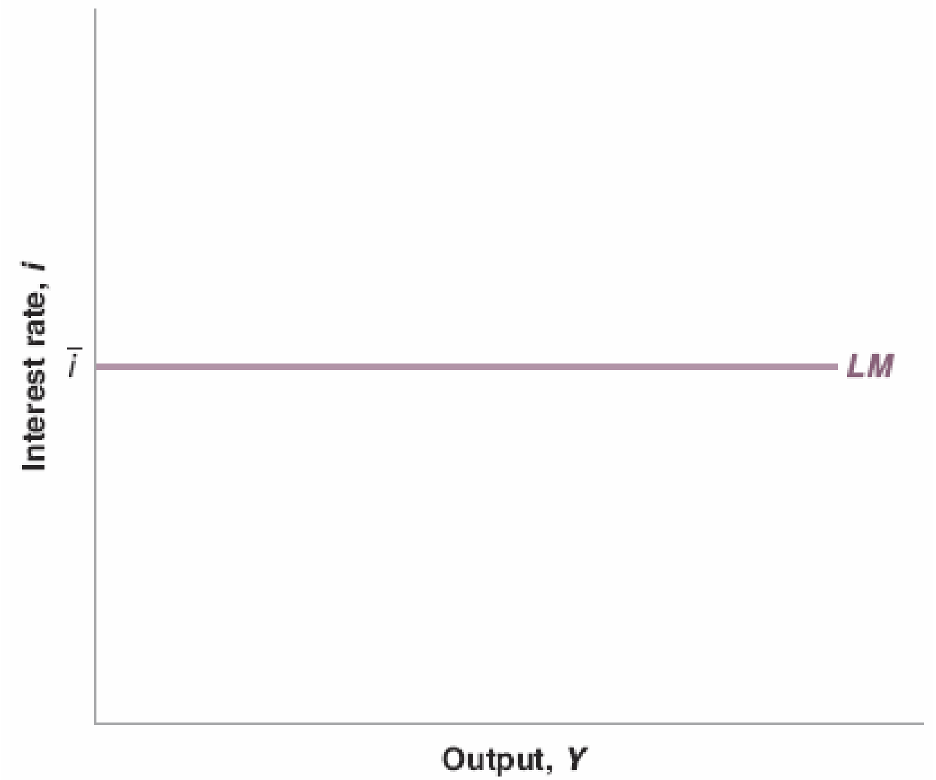
$$M=€YL(i)$$

- In terms of real demand of money and real income

$$M/P=YL(i)$$

- If a cappuccino costs 1.80 euros and I hold money for 2 cappuccinos a day, then
  - I hold **nominal money balances** equal to 3.60 euros
  - I hold **real money balances** equal to 2 cappuccinos

Deriving the LM curve



The central bank chooses the interest rate **and adjusts the money supply so as to achieve it, that is such that**

$$M = PYL(\bar{i})$$

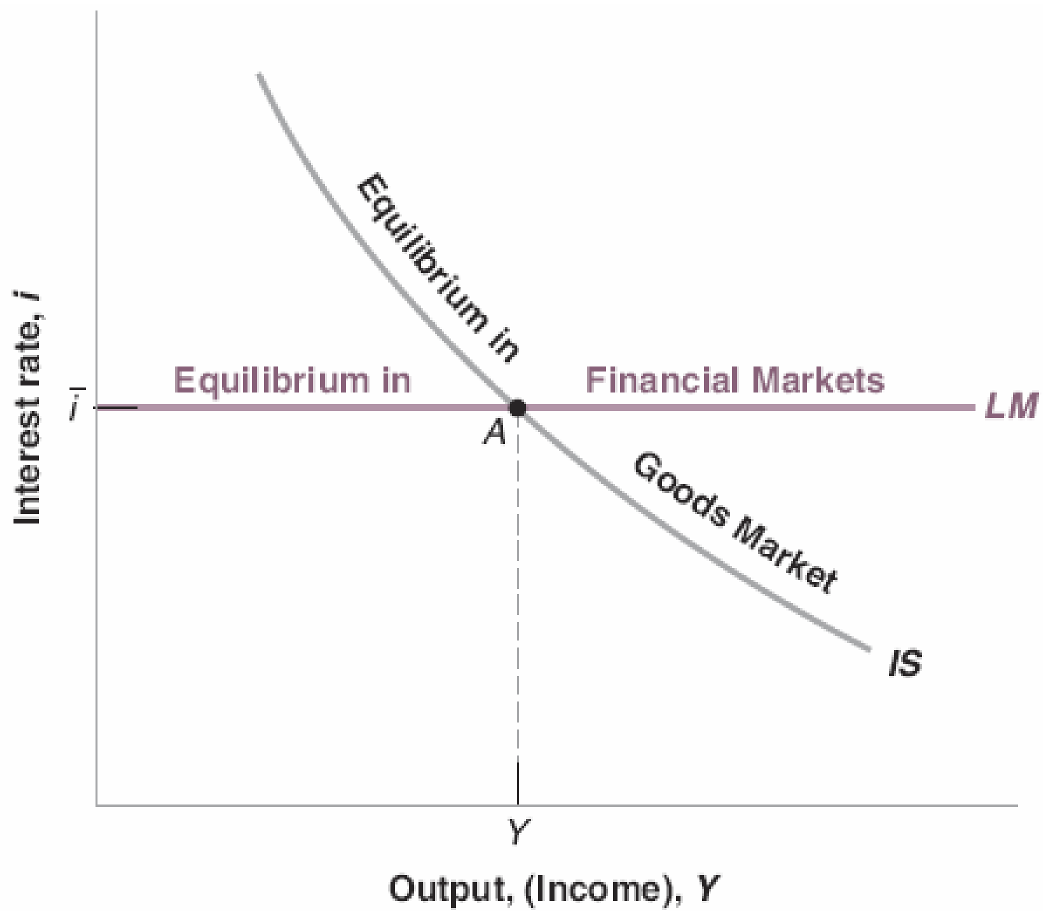
Let's put the IS and LM together!

$$\text{IS relation}: Y = C(Y-T) + I(Y,i) + G \qquad \text{LM relation}: i = \bar{i}$$

a) The **IS curve** represents the **equilibrium** in the **goods market.** It implies that an increase in the interest rate leads to a decrease in output

b) The **LM curve** represents the **equilibrium** in **financial markets.** The LM curve is horizontal at the target interest rate

c) At the **intersection** of the IS and LM curves **both goods** and **financial markets** are **in equilibrium**



Fiscal policy

1) A decrease in *G-T* is called a **fiscal contraction**, or **fiscal consolidation**

2) An increase in *G-T* is called a **fiscal expansion**

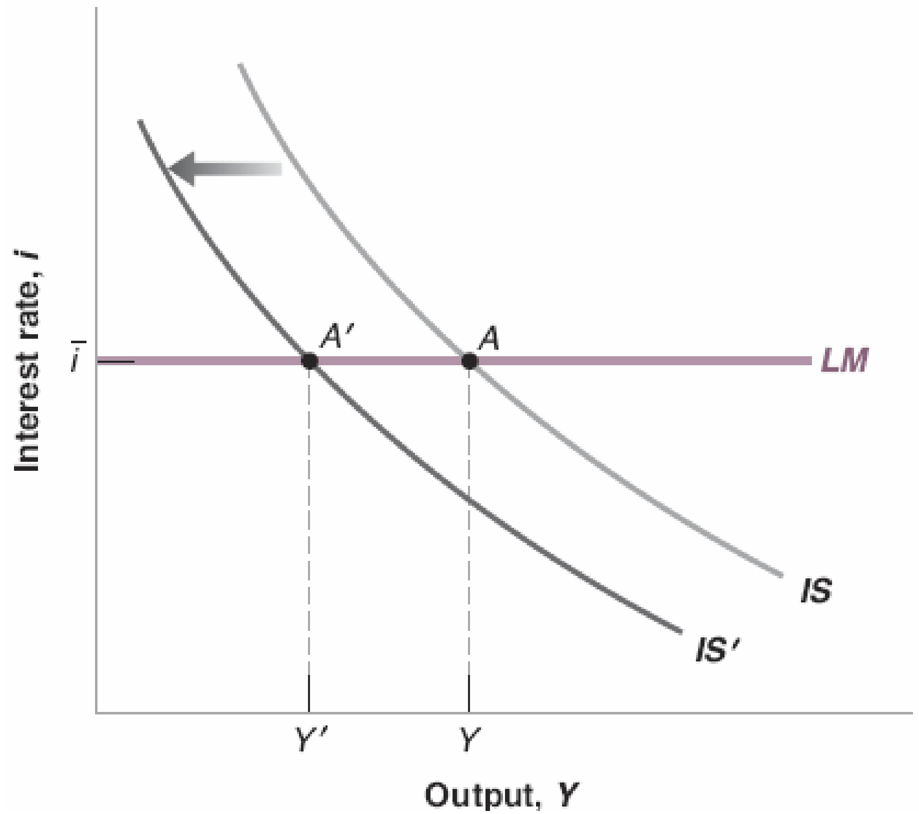
Fiscal policy: effects on output and interest rate

Example: a fiscal contraction, T increases

a) <u>Which curves moves?</u> LM does not move; IS shifts to the left

b) <u>What are the effects?</u> Output decreases; the interest rate does not change (by assumption on monetary policy)

c) <u>What is the economic intuition?</u> The increase in taxes leads to a decrease in disposable income, which causes a decrease in consumption. This decreases demand, output and income.



d) <u>What is the effect on composition of output?</u>

- The increase in *T* and the decrease in *Y* both reduce disposable income, so *C* decreases
- The decrease in *Y* leads to a decrease in *I*
- *G* has not changed

Monetary policy

1) A decrease in i (or an increase in M) is called a **monetary expansion**

2) An increase in i (or a decrease in M) is called a **monetary contraction**, or **monetary tightening**

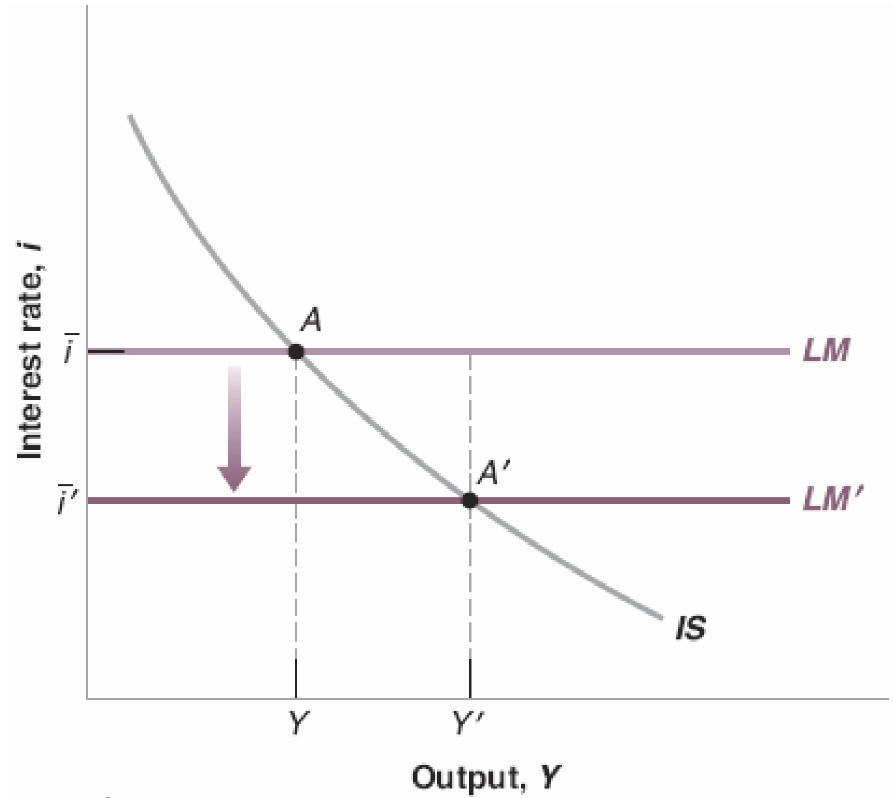
Monetary policy: effects on output and interest rate

Example: a monetary expansion, i decreases

a) <u>Which curves moves?</u> IS does not move; LM shifts down

b) <u>What are the effects?</u> Output increases; the interest rate decreases (by assumption on monetary policy)

c) <u>What is the economic intuition?</u> The decrease in the interest rate leads to an increase in investment, leading to an increase in demand, output and income



d) <u>What is the effect on composition of output?</u>

- The decrease in i and the increase in Y both increase I
- The increase in income Y leads to an increase in C
- G has not changed

Using a monetary-fiscal policy mix

For example when the economy is in a deep recession

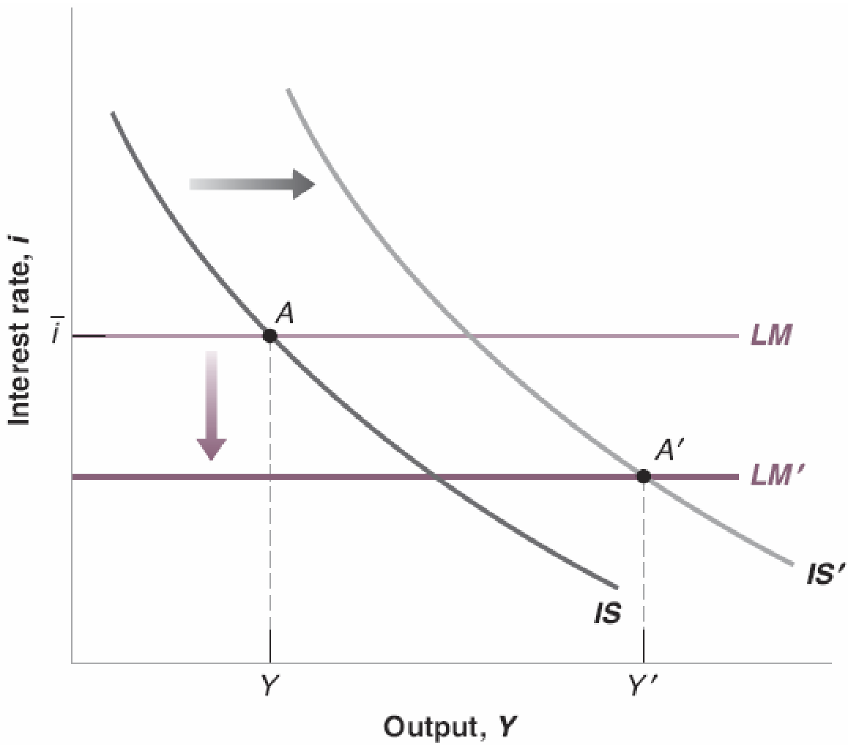
a) The economy is in a deep recession. It may be desirable to use fiscal and monetary policy in the same direction, say, a decrease in T and a decrease in i

b) Which curves moves? IS shifts to the right; LM shifts down

c) What are the effects? Output increases; the interest rate decreases

d) What is the effect on composition of output?

- The decrease in i and the increase in Y both lead to an increase in I
- The decrease in T and the increase in Y both lead to an increase in C
- G has not changed



Why a policy mix?

a) Why not just fiscal policy, say through a sufficiently large increase in G or decrease in T? Why not just monetary policy, though a sufficiently large decrease in the interest rate?

b) An expansionary fiscal policy means increasing the budget deficit (or decreasing the budget surplus). Running a large deficit or increasing government debt may be dangerous, as you will see later. It is better to rely at least in part on monetary policy

c) An expansionary monetary policy means decreasing the interest rate. If the rate is very low, then the room for monetary policy may be limited and fiscal policy may have to do more of the job. If the economy is at the *zero lower bound*, then fiscal policy has to do all the job

d) Fiscal and monetary policy have different effects on the composition of output:

- A decrease in taxes tends to increase consumption more relative to investment
- A decrease in the interest rate tends to increase investment more relative to consumption

Using a monetary-fiscal policy mix

For example when the government is running a large budget deficit

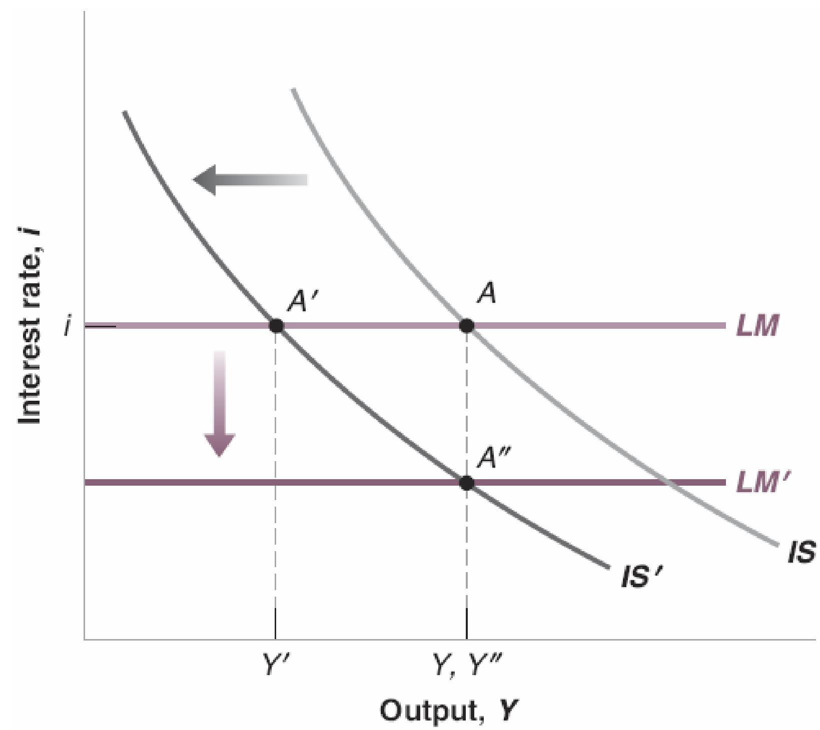
a) The government is running a large budget deficit and would like to reduce it, but does not want to trigger a recession. It may be desirable to use fiscal and monetary policy in the opposite direction, say, an increase in T and a decrease in i

b) Which curves moves? IS shifts to the left; LM shifts down

c) What are the effects? Output does not change, the interest rate decreases

d) What is the effect on composition of output?

- The decrease in i, at unchanged Y, leads to an increase in I
- The increase in T, at unchanged Y, leads to a decrease in C
- What would happen to C if the fiscal consolidation is obtained via a reduction in G?



Deficit reduction: good or bad for investment?

- Common argument: "*Private savings go either toward financing the budget deficit or financing investment. It does not take a genius to conclude that reducing the budget deficit leaves more savings available for investment, so investment increases*"

- Equilibrium in the goods market implies $I = S + (T - G)$

- Thus, given private saving (S), a lower public deficit (higher $T - G$) means higher I

- However, a fiscal contraction (if not accompanied by a reduction in interest rates) lowers output and so I decreases (since Y lower and i unchanged)

- This happens because as Y goes down, S also goes down. It goes down by more than $T - G$ increases, so total savings decrease and I decreases

- If instead the CB also lowers the interest rate to keep Y unchanged, then I necessarily goes up (since Y unchanged and i lower)

- **Conclusion**: whether a deficit reduction leads to a rise or a decrease in investment depends on the response of monetary policy

Let's do some math. Start with the IS

- The IS curve is

$$Y = C(Y - T) + I(Y, i) + G$$

- Assume everything is linear

$$C(Y - T) = c_0 + c_1(Y - T)$$

$$I(Y, i) = \bar{I} + d_1 Y - d_2 i$$

where **d_1 and d_2 are the sensitivities of *I* to Y and *i***

Combining we get

$$Y = \underbrace{[c_0 + c_1(Y - T)]}_{=C(Y-T)} + \underbrace{[\bar{I} + d_1 Y - d_2 i]}_{=I(Y,i)} + G$$

which can be rearranged, solving for Y, as

$$Y = \frac{1}{1-c_1-d_1}\underbrace{[c_0 + \bar{I} + G - c_1 T]}_{=\mathrm{A}} - \frac{d_2}{1-c_1-d_1} i$$

New goods market multiplier!

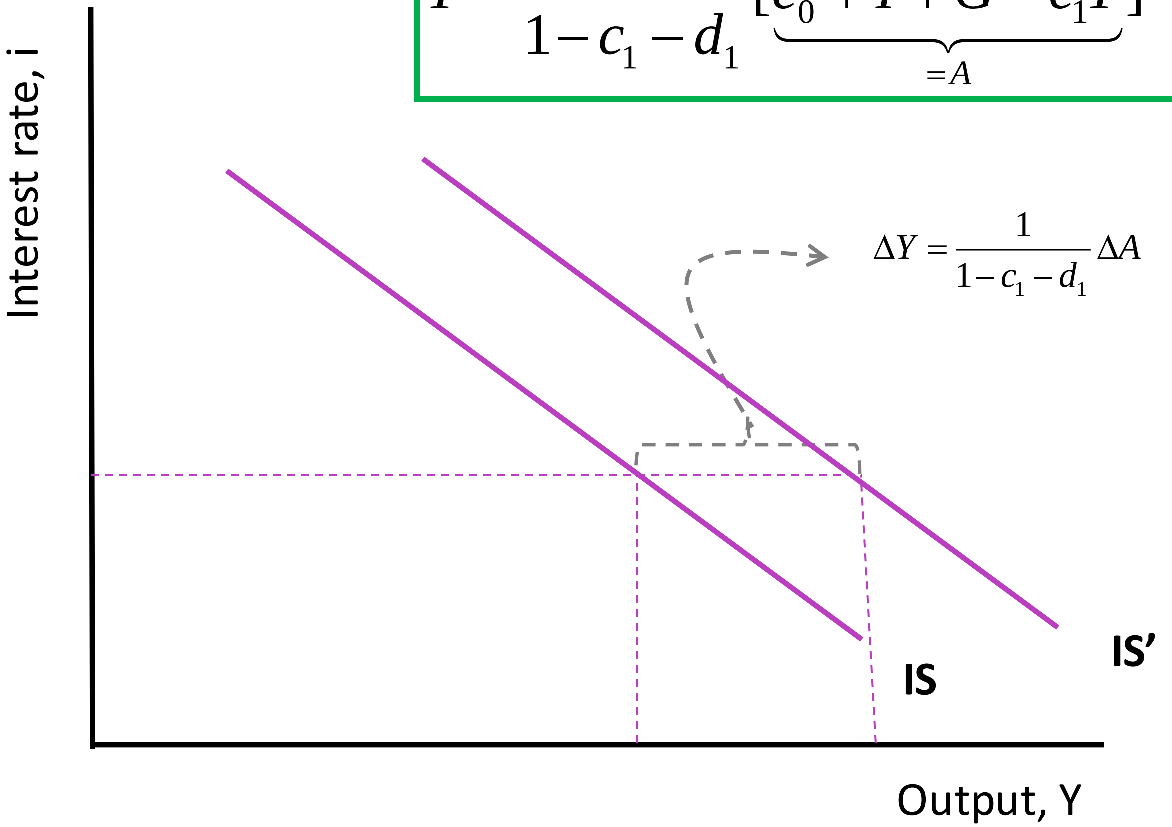
Autonomous spending

Effect of a change in i on Y

Note **d_2 is the sensitivity of demand to interest rates**; **$c_1 + d_1$ is the propensity to spend** (on consumption + investment)

Represent IS curve in a graph: position

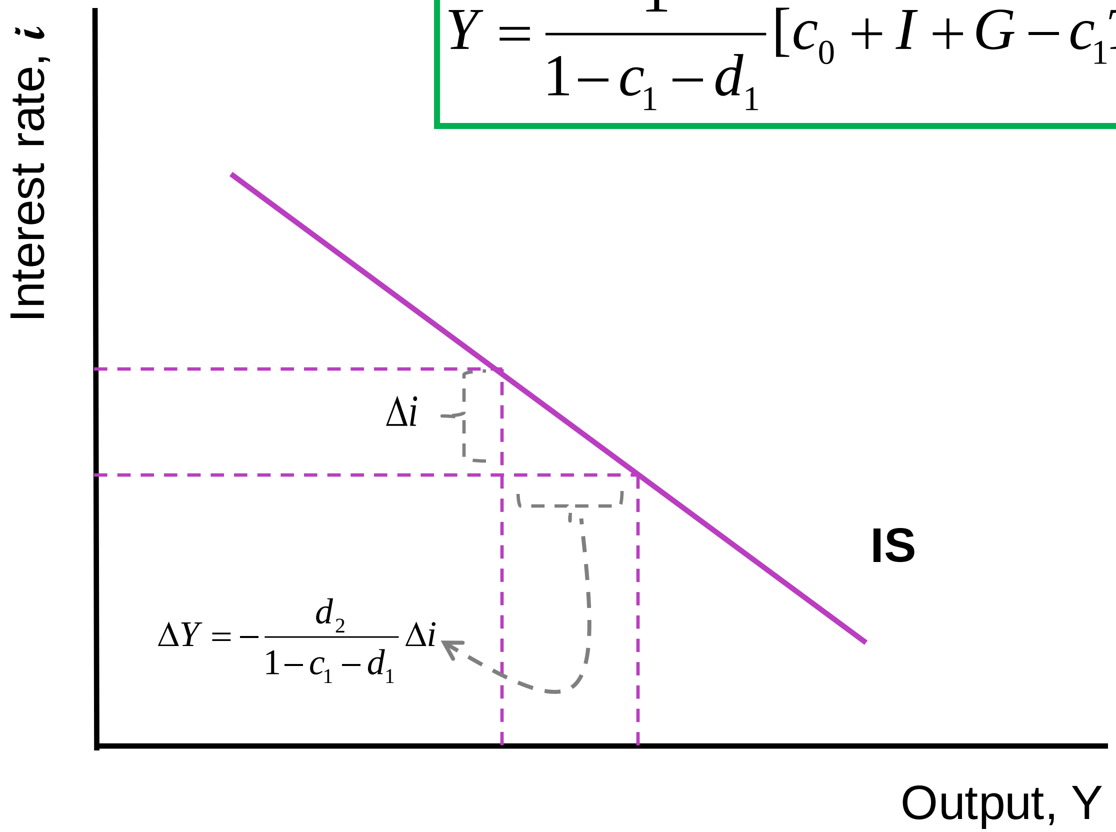
$$Y = \frac{1}{1-c_1-d_1}\underbrace{[c_0+\bar{I}+G-c_1T]}_{=A} - \frac{d_2}{1-c_1-d_1}i$$



Changes in any of the **components of autonomous spending A** will **shift** the **IS**

Represent IS curve in a graph: slope

$$Y = \frac{1}{1-c_1-d_1}[c_0 + \bar{I} + G - c_1 T] - \frac{d_2}{1-c_1-d_1} i$$



Changes in **c_1, d_1, and d_2** will change the **slope** of the IS

Turn to the LM curve

- Similarly to the IS we chose a functional form for the LM
- For convenience we rewrite the LM a

$$\frac{M}{P} = L(Y, i)$$

- And assume everything is linear

$$L(Y, i) = f_1 Y - f_2\, i$$

where **f_1 and f_2 are the sensitivities of real money demand to *Y* and *i***

- Combining

$$\frac{M}{P} = f_1 Y - f_2\, i$$

2 cases for LM:

Standard - endogenous money supply
Alternative - exogenous money supply

Endogenous money supply

- CB sets $i = \bar{i}$ and **adjusts the money supply *M*** according to:

$$M = P\left(f_1 Y - f_2 \bar{i}\right)$$

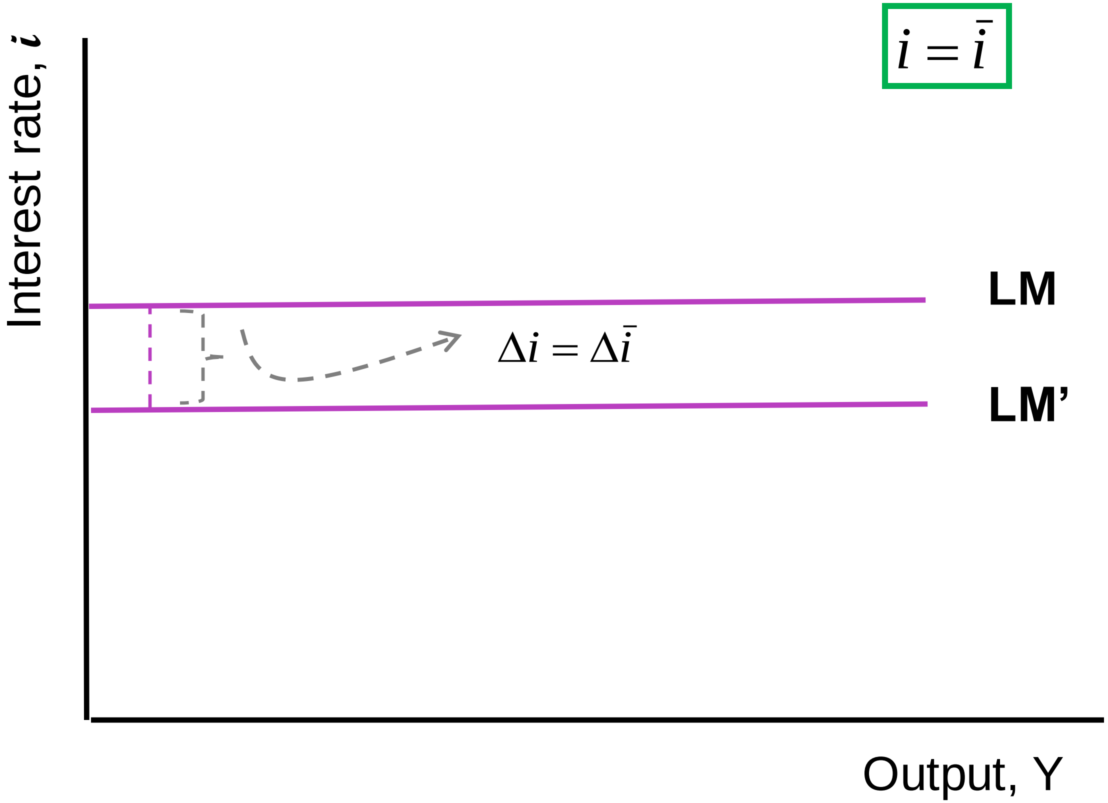
Exogenous money supply

- CB sets $M = \overline{M}$ and **let the interest *i* adjust** according to:

$$i = \frac{f_1}{f_2} Y - \frac{1}{f_2} \frac{\overline{M}}{P}$$

Standard case: endogenous money supply

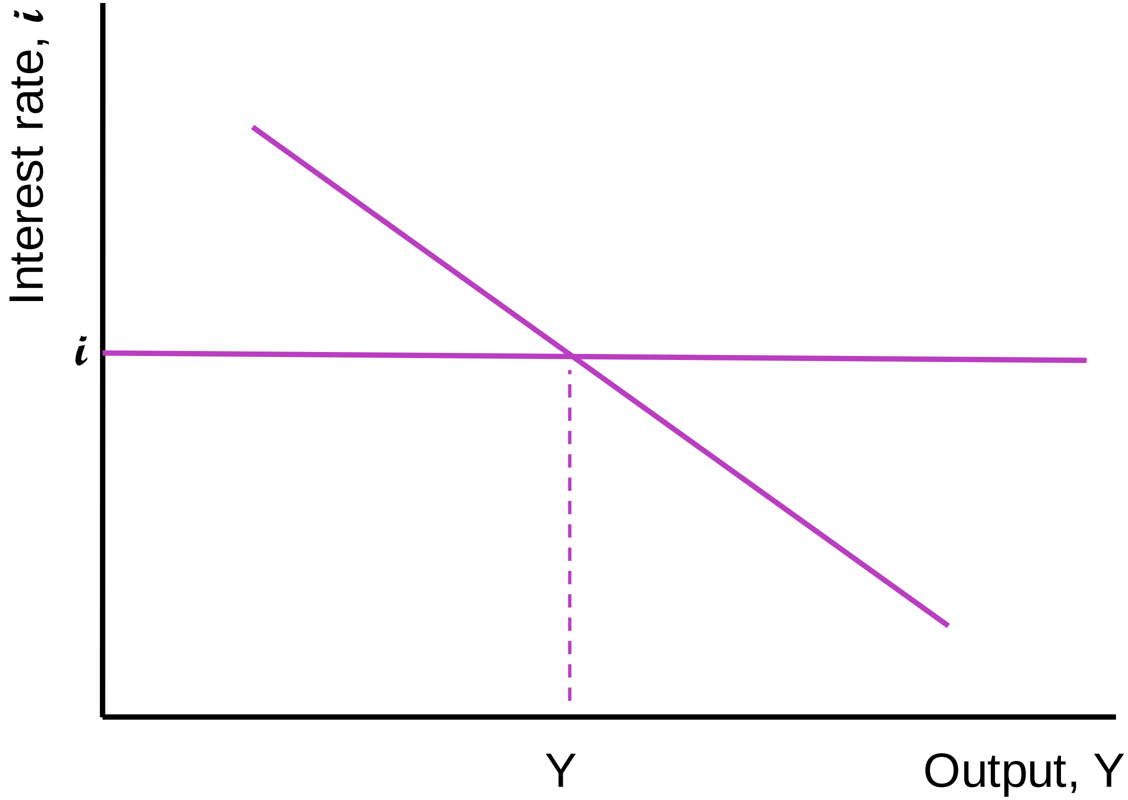
Represent LM curve in a graph: position and slope



Changes in the target interest rate shifts the LM curve. The slope is horizontal

Standard case: endogenous money supply

Equilibrium



Standard case: endogenous money supply

Equilibrium with math

IS

$$\boxed{Y = \frac{1}{1-c_1-d_1}[c_0+\bar{I}+G-c_1T]-\frac{d_2}{1-c_1-d_1}i}$$

LM

$$\boxed{i=\bar{i}}$$

This is a system in 2 equations and 2 unknowns. Which ones?

Solving the system we obtain **equilibrium output**

$$Y = \frac{1}{1-c_1-d_1}[c_0+\bar{I}+G-c_1T]+\frac{d_2}{1-c_1-d_1}\bar{i}$$

and **equilibrium interest rate**

$$i=\bar{i}$$

Consider **equilibrium output**

$$Y = \frac{1}{1-c_1-d_1}[c_0 + \bar{I} + G - c_1 T] + \frac{d_2}{1-c_1-d_1}\bar{i}$$

FISCAL POLICY MULTIPLIER

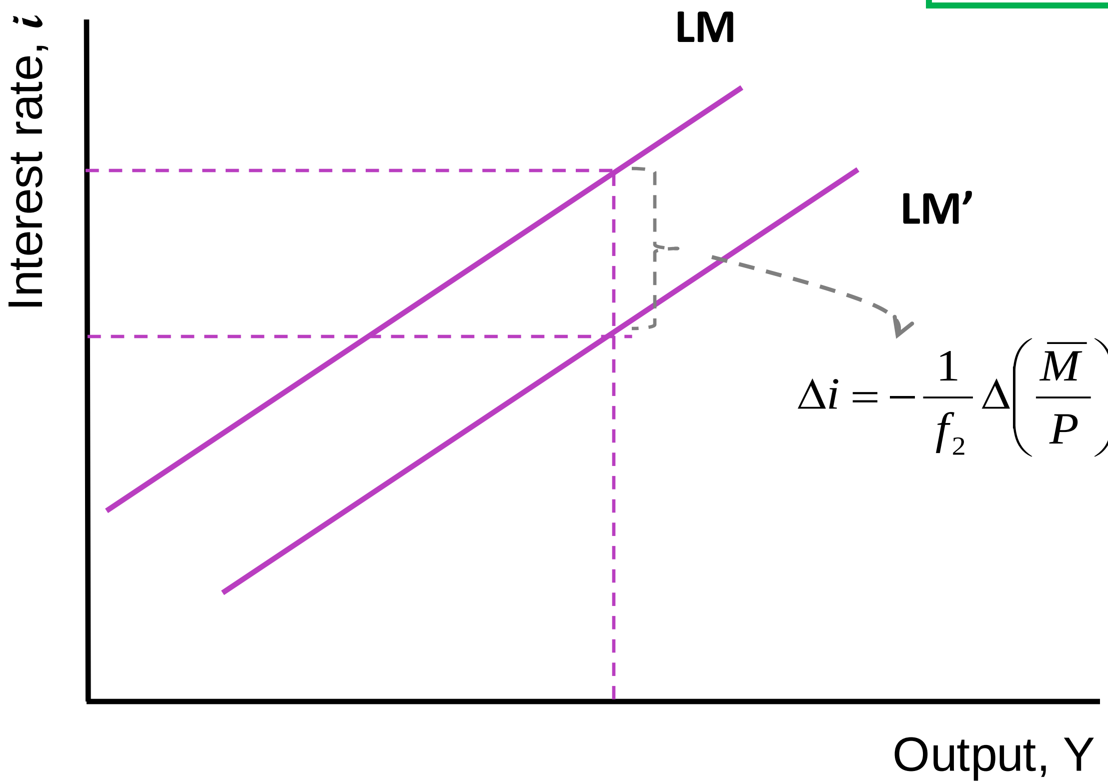
MONETARY POLICY MULTIPLIER

- The **fiscal policy multiplier** gives the **change in output caused by a change in *A*** (which includes changes in G and T)
- When the CB chooses the level of the interest rate (endogenous monetary policy), the fiscal policy multiplier is identical to the income multiplier derived studying the goods market in isolation
- The **monetary policy multiplier** gives the **change in output caused by a change in *i***
- The monetary policy multiplier equals 0 if d_2 equals 0, that is, if no component of aggregate demand depends on the interest rate

Alternative case: exogenous money supply

Represent LM curve in a graph: position

$$i = -\frac{1}{f_2}\frac{\overline{M}}{P} + \frac{f_1}{f_2}Y$$



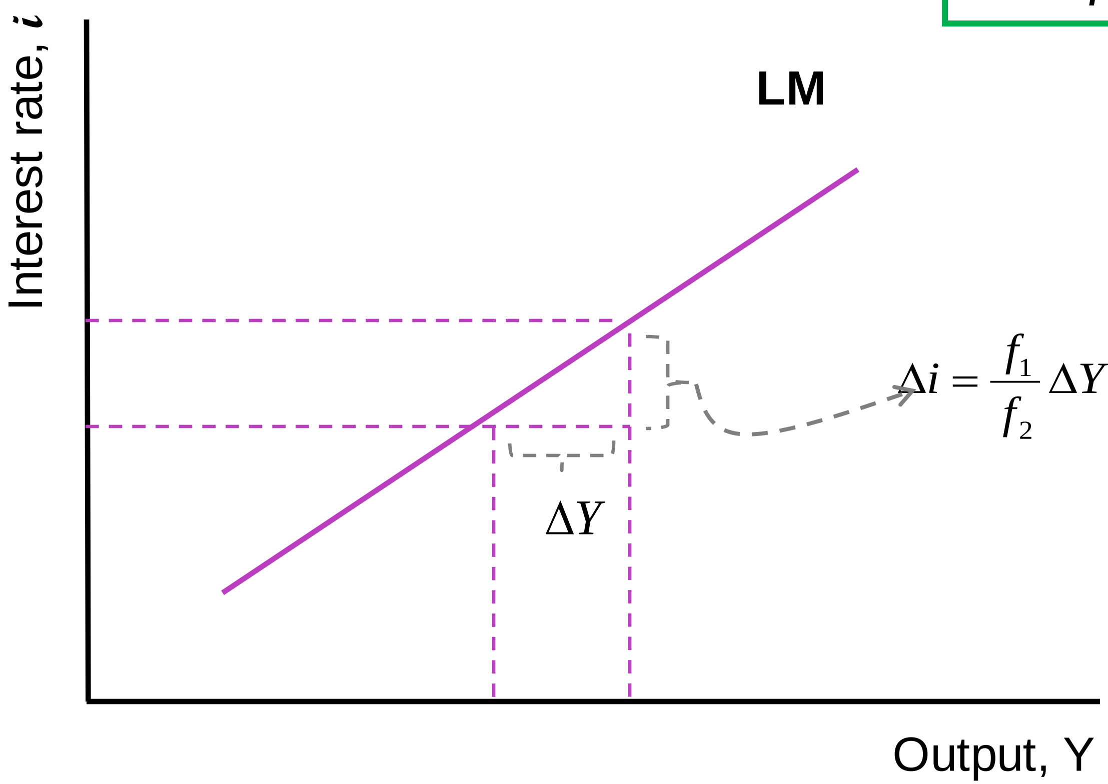
$$\Delta i = -\frac{1}{f_2}\Delta\left(\frac{\overline{M}}{P}\right)$$

Changes in **M** or **P** will **shift** the LM

Alternative case: exogenous money supply

Represent LM curve in a graph: slope

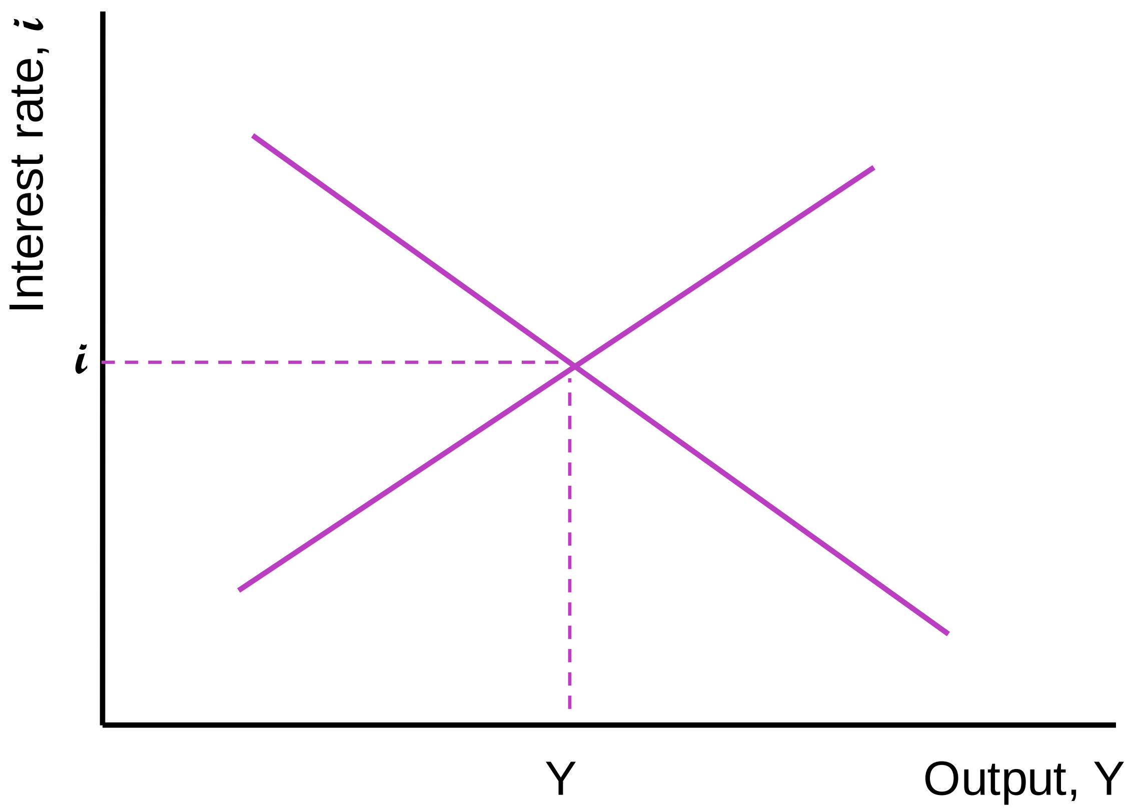
$$i = -\frac{1}{f_2}\frac{\overline{M}}{P} + \frac{f_1}{f_2}Y$$



Changes in **f_1 and f_2** will change the **slope** of the LM

Alternative case: exogenous money supply

Equilibrium



Alternative case: exogenous money supply

Equilibrium with math

IS

$$\boxed{Y=\frac{1}{1-c_1-d_1}[c_0+\bar{I}+G-c_1T]-\frac{d_2}{1-c_1-d_1}i}$$

LM

$$\boxed{i=-\frac{1}{f_2}\frac{\overline{M}}{P}+\frac{f_1}{f_2}Y}$$

- Again, this is a system in 2 equations and 2 unknowns, Y and i
- Now, however, money supply is exogenous
- Thus, we change the LM equation

Solving the system we obtain **equilibrium output**

$$Y = \frac{1}{(1-c_1-d_1)+\left(f_1 d_2 / f_2\right)}[c_0+\bar{I}+G-c_1 T]+\frac{d_2}{(1-c_1-d_1)f_2+f_1 d_2}\frac{\overline{M}}{P}$$

and **equilibrium interest rate**

$$i = \frac{f_1 / f_2}{(1-c_1-d_1)+f_1 d_2 / f_2}[c_0+\bar{I}+G-c_1 T]-\frac{1-c_1-d_1}{f_2\left(1-c_1-d_1\right)+f_1 d_2}\frac{\overline{M}}{P}$$

Consider **equilibrium output**

$$Y = \frac{1}{(1-c_1-d_1)+f_1 d_2/f_2}[c_0+\bar{I}+G-c_1 T]+\frac{d_2}{f_2(1-c_1-d_1)+f_1 d_2}\frac{\bar{M}}{P}$$

FISCAL POLICY MULTIPLIER

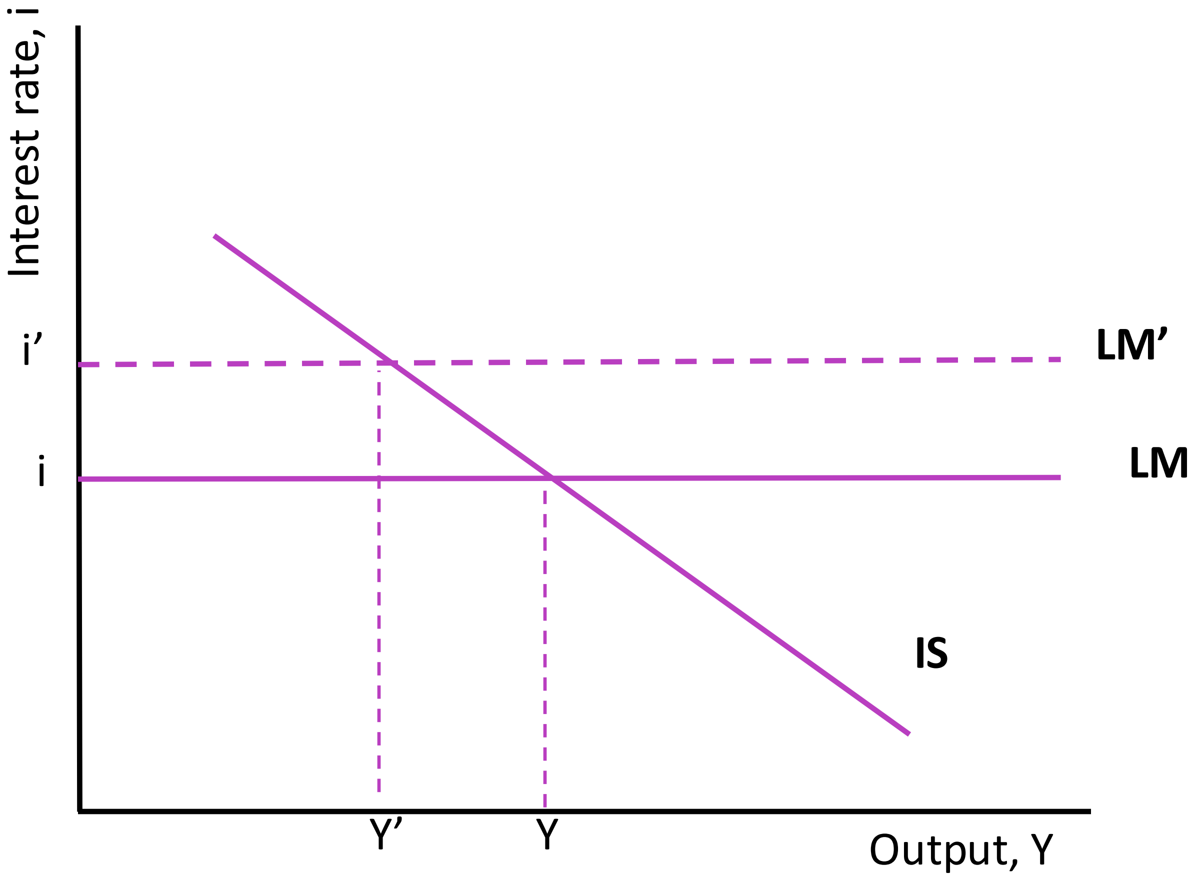
MONETARY POLICY MULTIPLIER

- The **fiscal policy multiplier** gives the **change in output caused by a change in A**
- It is smaller than the fiscal policy multiplier arising when the CB chooses the level of the interest rate
- The **monetary policy multiplier** gives the **change in output caused by a change in M (as opposed to i)**
- However, the effect of M still works through the effect on i, and ΔY/Δi is the same as before

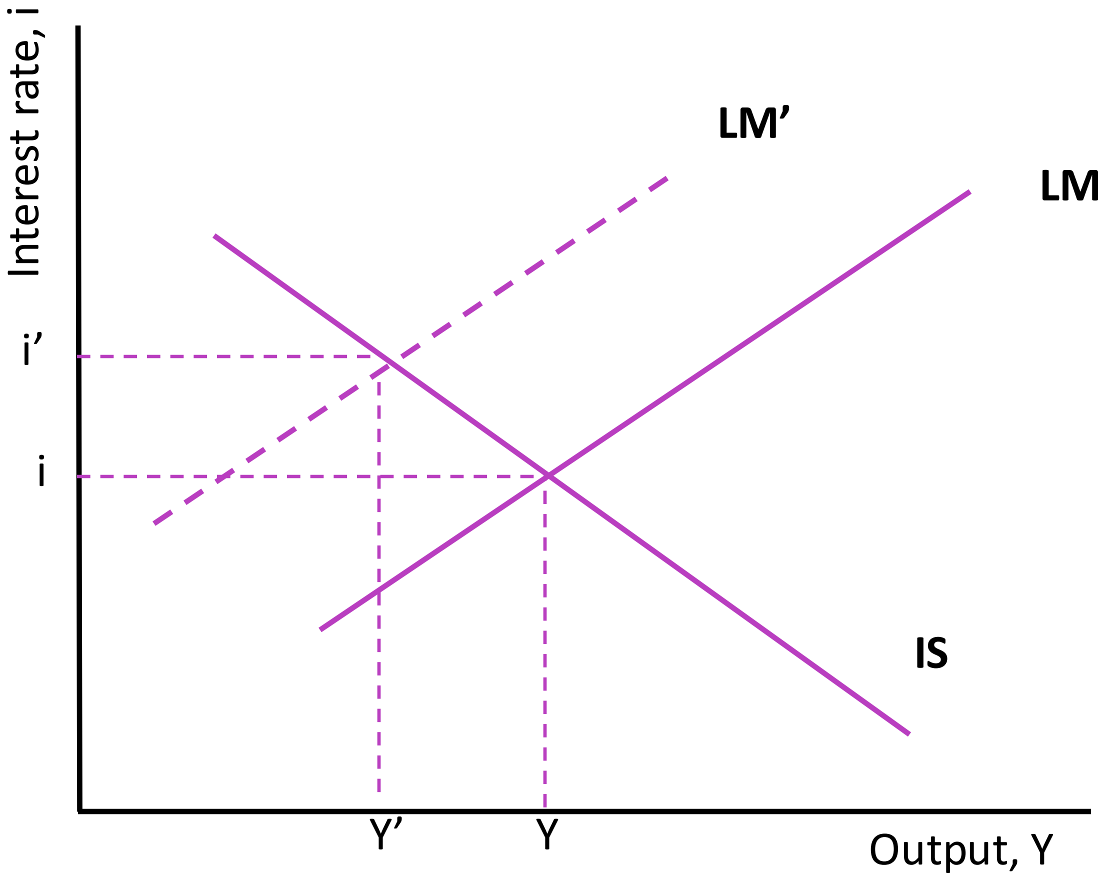
Back to the real world
How does the IS-LM model fits the facts?

- Suppose the CB implements a contractionary monetary policy: decreases M or increases i
- Recall, in the model the LM curve shifts up
- The IS-LM model predicts that **the interest rate increases** and **output decreases**
- What about prices? In the short run, prices moves little. For simplicity, IS-LM model assumes **constant prices**

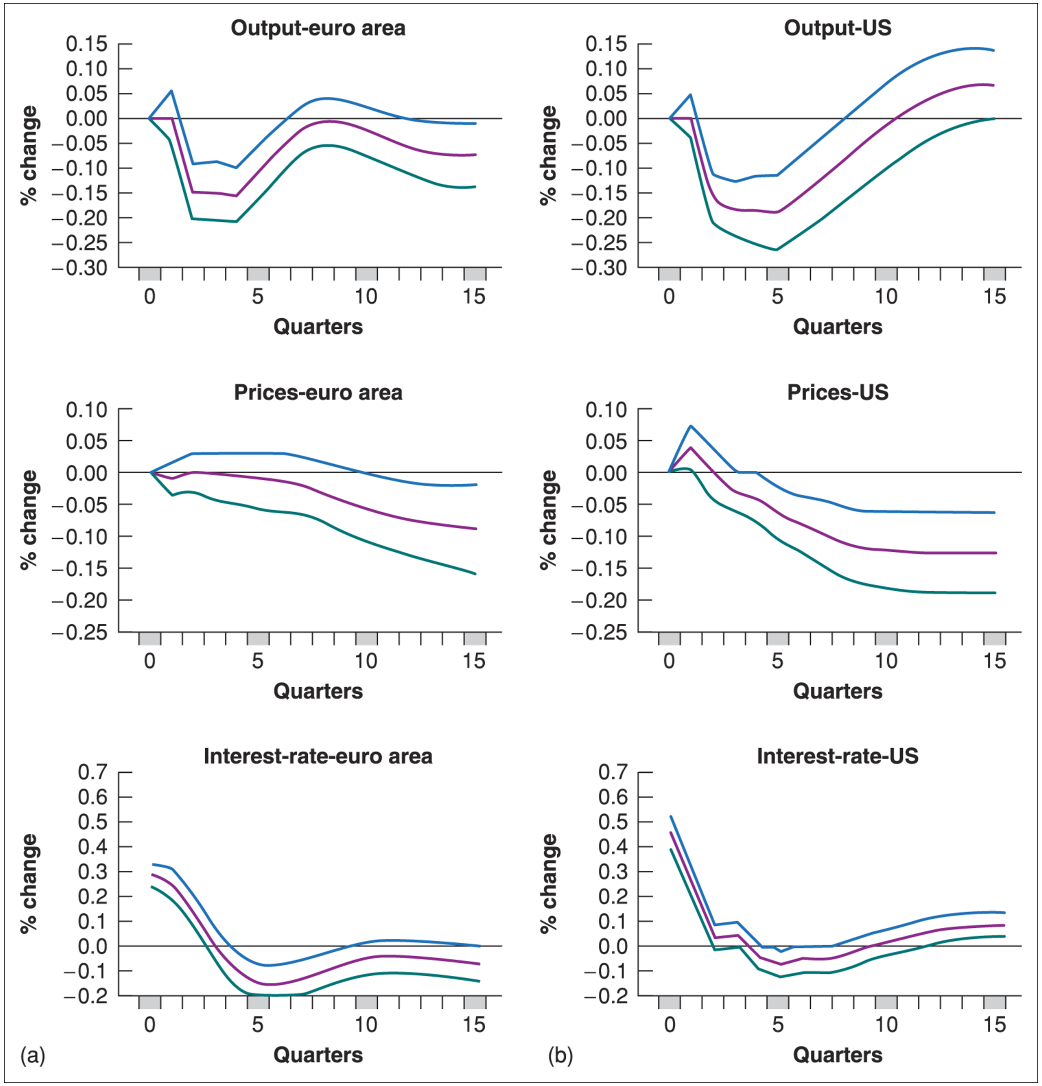
The IS-LM model predicts….



….or with exogenous money supply



The evidence is...



To understand the slow dynamics in previous evidence, consider that

- consumers are likely to take some time to adjust their consumption following a change in disposable income

- firms are likely to take some time to adjust investment spending following a change in their sales

- firms are likely to take some time to adjust investment spending following a change in the interest rate

- firms are likely to take some time to adjust production following a change in demand